

Adding fractions with different denominators

Unit 3 · Fraction operation · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A
Unit 3 + Modern Education 5A Unit 10

Core Trap: ☐ T9 Fraction Operations - After the common division, the numerator forgets to expand simultaneously · The result is not reduced

SSPA Association:  **High frequency** The score test is compulsory every year, accounting for about 15-20% of Paper 1

Prerequisite knowledge: Class 7 (Comparison of different denominators and common fractions · LCM calculation · Equivalent fractions and extended fractions)

Objectives of this class: ❶ Understand that different parts cannot be added directly ❷ Be proficient in the three-step method ❸ Addition of mixed numbers ❹ Application problems

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question · 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Calculate LCM(4, 6) = ?	Basic	
2	Calculate LCM(3, 5) = ?	Basic	
3	Divide $\frac{1}{2}$ and $\frac{1}{3}$ (write the two equivalent fractions after the common denominator)	Advanced	
4	Calculate $\frac{3}{8} + \frac{2}{8} = ?$ (Adding with the same denominator, P4 basics)	Basic	
5	Compare $\frac{2}{3}$ and $\frac{3}{5}$ which one is bigger? Write down the judgment process.	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Why can't fractions with different denominators be added directly? ●SSPA

- ① The denominator represents "the number of parts the whole is cut into." Different denominators = different sizes of each portion
- ② For example: $\frac{1}{2}$ is cut into **2 parts** · $\frac{1}{3}$ is cut into **3 parts**——The two parts are different sizes!
- ③ To add, it must first become **the same parts**→ This is the **common fraction**
- ④ The essence of the common fraction: use "equivalent fractions" (extended fractions) to change the denominator into LCM



✗ Cannot directly $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$

✓ Must be divided first: $\frac{1}{2} = \frac{3}{6}$, $\frac{1}{3} = \frac{2}{6} \rightarrow \frac{5}{6}$



□ **Example of trap detonation (the most important demonstration in this class)**

calculate: $\frac{5}{6} - \frac{1}{4} = ?$

✗ **Common mistakes (80% students)**

$$\frac{4}{2} = 2$$

Direct numerator minus numerator, denominator minus denominator

✓ **Correct solution**

$$\frac{7}{12}$$

$$\text{LCM}(6,4)=12 \rightarrow \frac{10}{12} - \frac{3}{12} = \frac{7}{12}$$

🧠 **Tip: "First understand the LCM, then multiply the numerator and keep the denominator unchanged. The answer must be simplified."**

⚠️ The most frequent error: adding and subtracting different denominators directly without common denominators. Remember: the denominators are different → be sure to connect the denominators first!

⚠️ The second most frequent error: only changing the denominator when dividing the whole number, forgetting to expand the numerator simultaneously. "The molecules follow and ride"!

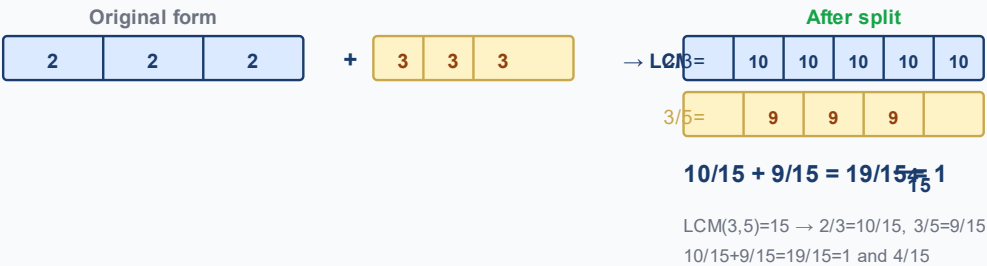
Knowledge Point - Worked Examples (write out ①LCM ②find a common denominator ③calculate ④simplify)

#	Question	Difficulty	Working Space
Example 1	$\frac{1}{4} + \frac{1}{2} = ?$	🌱	
Example 2	$\frac{1}{3} + \frac{1}{6} = ?$	🌱	

Knowledge point 2: General addition of fractions with different denominators (LCM is greater than the denominator itself) ● SSPA required test

Standard common three-step method:

① **LCM** ||||SEP|||: Short division to find the least common multiple ② **Common branch** ||||SEP|||: **simplest fraction/false** → **band**: Simultaneous expansion of molecules ③ **calculate**: Addition of numerators with the denominator unchanged ④ **reduce**: Simplest fraction/false → **band**



①

Looking for LCM

Find the lowest common multiple using short division

②

common points

The numerator and denominator change simultaneously to LCM

③

calculate

Adding numerators with the denominator unchanged

④

reduce

Simplest false fraction → mixed fraction

example

Example 3: $\frac{2}{3} + \frac{3}{5} = ?$

example

Example 4: $\frac{3}{8} + \frac{5}{6} = ?$

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
6	$\frac{1}{2} + \frac{1}{3} = ?$		
7	$\frac{2}{3} + \frac{1}{4} = ?$		
8	$\frac{3}{5} + \frac{1}{2} = ?$		

9	$\frac{3}{4} + \frac{1}{6} = ?$		
10	$\frac{5}{8} + \frac{1}{6} = ?$		

Knowledge Point 3: Addition of Different Denominators with Mixed Fractions

● SSPA Advanced

Recommended method (conversion to improper fraction method - the least error-prone):

- ① Mixed fraction → Improper fraction (integer × denominator + numerator) ② Common fraction (find LCM) ③ Calculation ④ Answer → Mixed fraction + reduction

$$\begin{array}{ccccccc}
 & \text{1 and 1/2} & & & \text{2 and 2/3} & & \\
 & \boxed{1} \boxed{1} & 1 & + & \boxed{2} \boxed{2} \boxed{2} & 2 & \\
 & \downarrow \text{Convert to false fraction} \downarrow & & & & & \\
 3/2 & + & 8/3 & \rightarrow & \text{LCM=6} & 9/6 + 16/6 & = 25/6 = 4 \frac{1}{6}
 \end{array}$$

1 and 1/2 + 2 and 2/3 → 3/2 + 8/3 → 9/6 + 16/6 = 25/6 = 4 and 1/6

example

Example 5: $1\frac{1}{2} + 2\frac{2}{3} = ?$

example

Example 6: $2\frac{1}{3} + 1\frac{1}{2} = ?$

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
11	$1\frac{1}{4} + 2\frac{2}{3} = ?$		
12	$1\frac{2}{5} + 2\frac{3}{10} = ?$		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
13	$\frac{1}{4} + \frac{1}{8} = ?$		
14	$\frac{1}{3} + \frac{1}{9} = ?$		
15	$\frac{1}{2} + \frac{1}{4} = ?$		
16	$\frac{1}{5} + \frac{3}{10} = ?$		
17	$\frac{2}{5} + \frac{1}{3} = ?$		

Advanced layer (total 5 questions,   choose do)

#	Question	Difficulty	Working Space
18	$\frac{3}{5} + \frac{1}{2} = ?$		
19	$\frac{5}{8} + \frac{1}{6} = ?$		
20	$\frac{3}{4} + \frac{1}{6} = ?$		
21	$1\frac{1}{4} + 2\frac{2}{3} = ?$		
22	$2\frac{1}{3} + 1\frac{1}{2} = ?$		

 challenge layer (total 3 questions,  choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
23	$2\frac{1}{3} + 1\frac{1}{2} + \frac{3}{4} = ?$ (Mixture of three fractions, including mixed numbers)		

24	$\frac{1}{2} + \frac{2}{3} + \frac{1}{6} = ?$		
25	A rope is divided into three sections: the first section is $\frac{1}{3}$ meters, the second section is $\frac{1}{4}$ meters longer than the first section, and the third section is $\frac{5}{6}$ meters. How many meters is the total length of the rope? (You need to find the length of the second paragraph first)		

Knowledge point 4: Application questions on addition of fractions (subject to sub-test word questions) SSPA compulsory exam

Four steps to solve the problem: ① Circle keywords ("Total" "Total" = addition) ② Column ③ Calculation of common score ④ Write a complete answer sentence (Step points will be deducted if you do not write an answer sentence!)

example

Example 7: Xiao Ming has $\frac{1}{2}$ Pizzas, and Xiaohua has $\frac{1}{3}$ Pizzas. How many pizzas do they have in total?

applicationquestionpractice (all must do, column → find a common denominator → calculate → answer sentence)

#	Question	Difficulty	Working Space
26	Xiao Mei drank $\frac{1}{4}$ liters of orange juice, and Xiao Ming drank $\frac{1}{2}$ liters. How many liters did they drink in total?		
27	A box of chocolate, my sister ate $\frac{2}{5}$ boxes, my brother ate $\frac{1}{3}$ boxes. How many boxes did the two of them eat in total?		
28	The original water bottle contains $\frac{3}{4}$ liters, then pour $\frac{1}{3}$ liters into it. How many liters are there now? (Answer with mixed fractions)		

advancedapplicationquestion ( choose do, SSPAPaper 2commonquestion type)

#	Question	Difficulty	Working Space
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29	The cake shop sold $\frac{2}{5}$ pieces in the morning and $\frac{1}{3}$ pieces in the afternoon. How many were sold in total throughout the day?		
30	The first section of rope is $1\frac{1}{4}$ meters, the second section is $2\frac{1}{3}$ meters. How many meters are the total length of the two sections? (Answer with fractions)		
31	SEP in the garden grows roses, $\frac{1}{4}$ grows chrysanthemums. What percentage of the garden is occupied by flowers? $\frac{2}{5}$ Plant chrysanthemums. What percentage of the garden is occupied by flowers?		
32	SEP in the granary is for white rice, $\frac{1}{6}$ is for brown rice, and $\frac{2}{5}$ is for red rice. What fraction of the granary is occupied by rice? What percentage is the remainder? $\frac{1}{3}$ Put red rice. What fraction of the granary is occupied by rice? What percentage is the remainder?		

IV. Ultimate challenge area

#	Question	Difficulty	Working Space
 1	$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} = ?$ (Adding four fractions with different denominators - find LCM(2,3,4,6))		
 2	A bottle of juice contains $1\frac{1}{2}$ liter. I drank $\frac{1}{3}$ liters for breakfast and $\frac{1}{4}$ liters for lunch. How many liters are left? (This class is learning addition, but this question uses subtraction!)		
 3	Normal points account for SEP , tests account for SEP , exams account for SEP . What percentage of the total score do the three items account for? What's left? (Total score = 1) $\frac{2}{5}$, the test accounts for $\frac{1}{3}$, the exam accounts for $\frac{1}{4}$. What percentage of the total score do the three items account for? What's left? (Total score = 1)		

V. Class afterhomework

Basic must-do questions (total 6 questions, must write find a common denominator step)

#	Question	Difficulty	Working Space
H1	$\frac{1}{4} + \frac{1}{8} = ?$		
H2	$\frac{1}{3} + \frac{1}{9} = ?$		
H3	$\frac{1}{2} + \frac{1}{6} = ?$		
H4	$\frac{2}{3} + \frac{1}{4} = ?$		
H5	$\frac{3}{4} + \frac{1}{6} = ?$		
H6	Xiao Ming has $\frac{1}{3}$ pieces of chocolate, and Xiaohua gives him $\frac{1}{2}$ pieces. How many lines does Xiao Ming have now?		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H7	A bottle of juice $\frac{3}{4}$ liters, then pour $\frac{1}{3}$ liters of water. How many liters are there now? (Answer with fractions)		
H8	Compare $\frac{5}{6} + \frac{1}{4}$ and $\frac{1}{2} + \frac{2}{3}$ which one is bigger? How much difference?		
H9	$1\frac{2}{3} + 2\frac{1}{2} + \frac{3}{4} = ?$		

VI. The Lessoncorecommon errorssummary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

✅ 本堂自我檢查 (完成後打剔)

☐ 我識得分辨每個知識點嘅陷阱

☐ 我能夠獨立完成 🧩 基礎題

☐ 我能夠挑戰 🌿 進階題

☐ 我記得住口訣

#	common error	Correct Approach
1	Add the different denominators directly : $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$	Different denominators → Be sure to connect the denominators first!
2	After splitting, the molecules forget to expand simultaneously.	"Multiply the numerator" - multiply the denominator by the same number and multiply the numerator by the same number
3	The answer is not reduced : $\frac{6}{12}$ when the answer	The final step must be to check the common factors
4	Improper fractions are not converted to mixed fractions	Numerator > Denominator → Convert into mixed numbers (subject to the requirements of the division test)
5	LCM error finding (use the larger common multiple)	Short division is confirmed to be the true "minimum"
6	Mixed fractions are added before they are converted to false	It is recommended to use the "improper fraction conversion method" which is the least error-prone
7	Missing sentences/units in application questions	Must write "Answer:..." + unit, otherwise step points will be deducted

🏠 家長角落 · Parent Corner

今日學咗咩? 小朋友學咗「**Adding fractions with different denominators**」。

最易錯嘅 3 個陷阱: 留意老師圈出嘅陷阱題目

你可以問小朋友:「你可唔可以解釋「**Adding fractions with different denominators**」嘅最重要口訣俾我聽?」

溫馨提示: 唔需要識教數學 — 只需要問小朋友「點解咁計?」同「有冇檢查陷阱?」就夠。

📄 教師參考: 熱身題答案 → 1)___ 2)___ 3)___ 4)___ 5)___ | 課堂練習重點關注題號: 🌿 挑戰層 17-21

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📖 Related topics: L07 Comparison of fractions with different denominators · L09 Multiplication of fractions · L22 Division of fractions